

AILT Study 6 (#152863)

Author(s)

Chiara Longoni (Bocconi University) - chiara.longoni2@unibocconi.it
Stephanie Tully (University of Southern California) - tullyste@marshall.usc.edu
Gil Appel (George Washington University) - gappel@gwu.edu

Pre-registered on: 2023/11/28 - 10:04 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining whether AI literacy impacts receptivity toward AI, and whether any such relationship is explained by perceptions of AI as being more magical and eliciting lower feelings of awe. Specifically, we intend to examine whether increased AI literacy leads to lower perceptions of AI as being magical, which in turn leads to lower feelings of awe, and hence lower receptivity towards AI.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent measure is receptivity toward AI as measured by responses on the following four items taken from IPSOS global AI receptivity survey (<https://www.ipsos.com/en/global-opinions-about-ai-january-2022>): I believe that products and services using artificial intelligence will profoundly change my daily life in the next 3-5 years; I believe that products and services using artificial intelligence have more benefits than drawbacks; I believe that products and services using artificial intelligence make my life easier; I trust companies using artificial intelligence as much as I trust other companies. All will be measured on 7-point scales (1 = not at all, 7 = very much)

The mediator capturing beliefs of AI as magical will be measured with participants' level of agreement with the following six statements: Artificial intelligence seems magical; Artificial intelligence seems to conjure answers out of thin air; Artificial intelligence seems like wizardry; Artificial intelligence seems superhuman; Artificial intelligence has an air of mystique; If I could see how artificial intelligence works, it would look like a magic show. All will be measured on 7-point scales (1 = strongly disagree, 7 = strongly agree)

The mediator capturing feelings of awe toward AI will be measured with participants' level of agreement with the following four statements (taken from Koutish et al 2023): "If I think about artificial intelligence that powers products and services: I experience feelings of awe; I feel that I am witness to something grand; I perceive vastness and I feel my jaw drop; I have goosebumps and gasp. All will be measured on 7-point scales (1 = strongly disagree, 7 = strongly agree)

4) How many and which conditions will participants be assigned to?

There will be no conditions. Our independent variable will be participants' AI literacy as measured using a 17-item multiple choice test. The sum of the correct responses will serve as their AI literacy level.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' receptivity toward AI across the 4 items. We will then regress this average AI receptivity on their AI literacy score. We will calculate participants' perception of AI as magical by averaging across the 6 measures and participants' feelings of awe towards AI across the 4 items.

We will examine serial mediation using Hayes' PROCESS macro 6 with AI literacy as the IV, receptivity as the DV, magic as the first mediator and awe as the second mediator.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include one instructional manipulation check (IMC; "Please indicate you are taking the time to read the instructions carefully. This is not a trick question. What is 12-4? Please use a numeric value only (no letters)."). This will be asked at the beginning of the survey and participants who fail this question will not complete the survey.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We aim to recruit 1,000 participants using the platform Prolific.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will collect perceptions of capability of AI by assessing the extent to which participants think that an AI algorithm is capable of (1=definitely incapable, 7=definitely capable): Composing a song; Recommending a gift; Playing the piano; Driving a car; Recommending music; Piloting a plane; Writing a news article. We will average responses to these items into an index capturing capability perceptions.

We will collect fear of AI by assessing the extent to which participants agree with the following four statements: I fear that artificial intelligence will make us lose our humanity; I fear that artificial intelligence will turn against humans; I fear that artificial intelligence will make humans become obsolete; I fear that artificial intelligence will make humans lose the ability to connect with each other. All will be measured on 7-point scales (1 = not at all, 7 = very much). We will average responses to these items into an index capturing fear of AI.

We will also measure demographic information (gender, age, income).